MINI-PROJECT SHORT TECHNICAL REPORT

Course: Cross-Platform Mobile App Development (VKU)

Mini-Project Title: Mini-Project 2: VKU Room Booking & Real-Time QR Check-In System

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1. GENERAL INFORMATION & DELIVERABLE LINKS

* **Team Members:**

1. Nguyễn Thị Thương — Student ID: 23IT.B219 — Lớp SH: 23SE4 — Lớp HP: Phát triển ứng dụng di động đa nền tảng (4) — Role: Full-Stack Developer — Contribution: 100%

* 🔗 **Live Demo URL (VKU Room Booking):** https://vku-room-booking-17t.pages.dev/
* 🔗 **Linked Ecosystem App (VKU Field Survey - Mini-Project 1):** https://vku-field-survey-28x.pages.dev/
* 💻 **GitHub Repository:** https://github.com/thuong614oh65/vku-room-booking-expo
* 📱 **APK Download Link:** app-release.apk (78.8 MB)
* 🎥 **Video Demo (Optional):** N/A

2. FEATURE IMPLEMENTATION CHECKLIST

#	Required Feature	Status	Implementation Details & Acceptance Level
1	Responsive Mobile Viewport	✅ Complete	100% responsive across mobile viewports via React Native Web; dark navy header, bottom tab navigation, custom VKU branding icons (`192px` , `512px` , `1024px`).
2	Real Google OAuth 2.0 (`@vku.udn.vn`) & `SHA-256` Auth	✅ Complete	Official Google Identity Services (`accounts.google.com/gsi/client`) with `hd: vku.udn.vn` domain restriction + `SHA-256 + Salt` local password hashing (`authService.ts`).
3	Auto Zalo WebView → Safari/Chrome Redirect	✅ Complete	Detects embedded WebViews (`Zalo` , `Messenger` , `FBAN`) and auto-redirects to **Safari** on iOS (`x-safari-https://`) or **Chrome** on Android (`intent://`) so Google OAuth & Camera work seamlessly.
4	Local Offline Persistence	✅ Complete	Uses Zustand + `@react-native-async-storage/async-storage` for local caching of bookings, waitlist, and authenticated student profile.
5	Automatic Background Sync	✅ Complete	Auto-syncs booking data to Google Cloud Firestore & `ntfy.sh` pub/sub cloud channels upon network reconnection (`isSynced` flag queue).
6	Multi-Parameter Filter	✅ Complete	Real-time filtering by keyword, building (Khu A/B/C/V/Lib), capacity, equipment, and date/time slot.
7	VKU 2-Hour Slot Grid	✅ Complete	5 standard VKU time slots (07:30–19:30) with visual locking for occupied slots in `TimeSlotGrid.tsx` .
8	First-Committed-Wins Booking	✅ Complete	`atomicAddBooking()` in Zustand store prevents duplicate bookings via synchronous `checkSlotConflict()` .
9	Smart Conflict Resolution	✅ Complete	Modal recommends alternative rooms (±10 seats), adjacent open slots, or priority waitlist entry.
10	Priority Waitlist	✅ Complete	Students queue for booked slots; push notification dispatched automatically when slot is released.

#	Required Feature	Status	Implementation Details & Acceptance Level
11	Digital Booking Pass & Smart Verification QR	✅ Complete	Electronic pass with Student ID, room code, slot time, UUID, and scannable URL QR matrix (`/verify/?id=...`).
12	Universal QR Check-In (Zalo / Phone Camera / In-App)	✅ Complete	Scanning via phone Camera/Zalo opens `/verify/index.html` and broadcasts real-time `CHECKED_IN` status (`< 0.5s` via `ntfy.sh` SSE + Polling) to PC/Phone; also supports in-app `jsQR` camera & image upload.
13	Cross-App Integration with VKU Field Survey (MP1)	✅ Complete	One-tap `📊 Đánh Giá CSVC (VKU Field Survey)` button passes `room` , `building` , `floor` , `name` , and `studentId` via URL query params to Mini-Project 1 (`vku-field-survey-28x.pages.dev`) with real-time server clock.
14	Booking Lifecycle Management	✅ Complete	`MyBookingsScreen` with tabs (Upcoming / Checked-In / Cancelled) and one-tap cancellation.
15	15-Minute Reminder Notification	✅ Complete	Background scheduler sends push alert 15 minutes before check-in time.
16	Cloud Firestore 2-Way Sync	✅ Complete	Real-time bidirectional sync via Google Cloud Firestore REST API for multi-device consistency.
17	Native Android APK	✅ Complete	Production APK (78.8 MB) built via Expo EAS Build, installable on Android 10–15 devices.
18	PWA Standalone Mode	✅ Complete	Full PWA with custom VKU Room Booking icons (`192x192` , `512x512` , `180x180`), Service Worker v15, and install prompt.

3. TECHNICAL ARCHITECTURE & PROJECT STRUCTURE

* **Directory structure:** App root `App.tsx` → `src/screens/` (BrowseRooms, RoomDetails, BookingConfirmationPass, MyBookings, QRScanner, Profile) → `src/components/` (AuthModal, RoomCard, FilterBar, TimeSlotGrid, ConflictResolutionModal, QRCodeModal, InstallAppModal) → `src/hooks/useRoomsQuery.ts` (TanStack Query) → `src/store/useBookingStore.ts` (Zustand) → `src/services/` (`authService.ts` , `googleOAuthPopupService.ts` , `cloudCheckInSyncService.ts` , `qrVerificationLink.ts` , `fieldSurveyLinkService.ts` , `notificationService.ts` , `realtimeBookingService.ts`) → `public/` (`manifest.json` , `sw.js` , `verify/index.html` , `icons/`).

* **State management & real-time sync flow:** Hybrid State Architecture combining **TanStack Query** (`@tanstack/react-query`) for server state/room caching and **Zustand** (`useBookingStore.ts`) for client state, session, and atomic booking transactions. Cross-device QR Check-In synchronization is powered by `cloudCheckInSyncService.ts` using **Server-Sent Events (`EventSource` SSE)** + HTTP polling over `ntfy.sh` plus `BroadcastChannel` for instant `< 0.5s` status updates between Phone and PC.

* **Exception handling:** (1) Concurrent booking collisions handled by synchronous pre-flight check before state mutation. (2) Embedded WebView restrictions (`403 disallowed_useragent` on Zalo/Messenger): automatically escaped via `x-safari-https://` (iOS) and `intent://` (Android Chrome). (3) Camera permission denied or desktop environment: QR scanner provides graceful fallback to upload a QR image/screenshot (`jsQR` canvas decoder).

4. EMPIRICAL EVIDENCE & SCREENSHOTS

Screenshot 1 — Browse Rooms & Header Integration:* Dark navy header with custom VKU Room Booking logo, `🟢 Online` / `⚡ Real-time` badges, `📊 Đánh Giá CSVC` cross-app button, building filter chips (Khu V, A, B, C, Lib), and 60 FPS `FlatList` .

Screenshot 2 — Real Google OAuth 2.0 (`@vku.udn.vn`) & Zalo Auto-Redirect:* Official Google Sign-In popup enforcing `@vku.udn.vn` student accounts and auto- extracting MSSV/Class, with automatic Zalo WebView → Safari redirect on iPhone.

Screenshot 3 — Digital Booking Pass & Cross-Device QR Check-In:* Student pass (`Nguyễn Thị Thương / 23IT.B219 / 23SE4 / V.A201`) with dynamic QR code; scanning on phone opens `/verify/index.html` receipt and immediately updates the booking pass to ✅ `ĐÃ CHECK-IN THÀNH CÔNG` .

Screenshot 4 — Ecosystem Link to VKU Field Survey (Mini-Project 1):* Clicking 📊 **`Đánh Giá CSVC Phòng V.A201`** opens `https://vku-field-survey-28x.pages.dev/` with pre-filled room code, building, floor, student name, MSSV, and live server-calibrated timestamp.

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5. TECHNICAL CHALLENGES & RESOLUTIONS

Challenge 1 — Concurrent Booking Race Condition: *When two students book the same slot simultaneously, async writes can cause duplicates.* Resolution:* Implemented synchronous `checkSlotConflict(roomId, date, slotId)` inside `atomicAddBooking()` so only the first commit is confirmed; the second triggers `ConflictResolutionModal` with alternative rooms and slots.

Challenge 2 — Google OAuth ***`403 disallowed_useragent`*** ***in Zalo WebView:*** *Google blocks OAuth 2.0 inside embedded in-app browsers (Zalo, Messenger) for security.* Resolution:* Removed all simulated in-app login screens, integrated official Google Identity Services (`accounts.google.com/gsi/client`), and implemented automatic WebView detection in `index.html` , `verify/index.html` , and `googleOAuthPopupService.ts` that immediately triggers `x-safari-https://...` on iOS or `intent://...#Intent;package=com.android.chrome;end` on Android to open the native browser.

Challenge 3 — Cross-Device QR Check-In Without Same-Browser Session: *Scanning a QR code generated on a laptop using a phone's native Camera or Zalo runs on two completely separate devices.* Resolution:* Encoded booking metadata into a signed verification URL (`/verify/?id=...`) that dispatches an instant cloud pub/sub payload via `ntfy.sh` (`cloudCheckInSyncService.ts`), updating the laptop/phone app screen to `CHECKED_IN` in under `0.5s` .